InboxRL: Reinforcement Learning via GRPO for Verifiable Calendar Event Extraction from Unstructured Academic Emails

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GitHub Repository: https://github.com/hhemantpatel/inboxrl

***Abstract*—Extracting structured calendar schedules from unstructured institutional correspondence is a high-value real-world automation task. Traditional supervised fine-tuning (SFT) suffers from exposure bias, hallucinated dates, and fragile formatting when encountering complex forwarded threads or relative dates. In this work, we present InboxRL, an end-to-end Reinforcement Learning system that trains a small language model (Qwen2.5-0.5B-Instruct) *purely via RL* without any supervised warm-start. We formulate structured JSON generation as a sequential Markov Decision Process (MDP) optimized using Group Relative Policy Optimization (GRPO). Terminal rewards are computed objectively through bipartite Hungarian matching across date, time, title, and venue fields, alongside explicit format and empty-state penalties to prevent reward hacking. Evaluated on synthetic and held-out anonymized real institute emails, InboxRL achieves a 96.8% valid JSON rate, an event F1 score of 88.4%, and 93.1% date accuracy, outperforming 3-shot prompting and heuristic baselines while running efficiently within 4 GB VRAM.**

***Index Terms*—Reinforcement Learning, GRPO, Small Language Models, Verifiable Rewards, Hungarian Matching, Information Extraction.**

I. Introduction

Students and faculty in academic institutes receive scores of unstructured emails daily announcing assignment deadlines, exam timetables, compensatory lectures, and seminars. Manually synchronizing these announcements into personal calendars (such as Google Calendar or iCal) is tedious and prone to missed deadlines.

While large language models (LLMs) can extract schedules via few-shot prompting, smaller edge models suitable for local, private deployment on student laptops frequently fail on formatting constraints, hallucinate spurious dates, or stumble on relative temporal references (e.g., "next Tuesday" relative to email receipt). Supervised fine-tuning (SFT) addresses some format errors but provides no policy feedback when intermediate token choices drift off distribution.

In accordance with strict course rules barring supervised and heuristic approximations, we build **InboxRL**: a pure RL policy trained using Group Relative Policy Optimization (**GRPO**) [1]. Ground-truth labels are utilized solely inside a non-differentiable terminal reward function, never as teacher-forced token targets.

II. MDP Formulation

We formulate sequence generation as an episodic, finite-horizon Markov Decision Process: $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$.

A. State and Action Space

- **State** $\mathbf{s}_t \in \mathcal{S}$: Represents the prompt context and token history up to step t:

$$s_t = (x_{\text{sys}}, x_{\text{email}}, d_0, y_1, y_2, \ldots, y_{t-1}) \tag{1}$$

  where x_{sys} is the system instructions, x_{email} is the email headers and body, d_0 is the reference received date (YYYY-MM-DD), and $y_{<t}$ are generated tokens. In Qwen2.5-0.5B [2], states are represented by 896-dimensional hidden representations over a max context length of $512 + 192 = 704$ tokens.
- **Action Space** $\mathbf{a}_t \in \mathcal{A}$: The discrete vocabulary space $\mathcal{V}$, where $|\mathcal{V}| = 151,936$ tokens.
- **Transition** $\mathcal{P}(s_{t+1}|s_t, a_t)$: Deterministic token append transition: $s_{t+1} = s_t \circ a_t$. An episode terminates when $a_t = \langle |\text{im_end}| \rangle$ or token count reaches $T = 192$.
- **Discount Factor**: $\gamma = 1.0$ (undiscounted episodic task).

B. Verifiable Reward Function

Intermediate actions yield $R_t = 0$. At episode termination, the decoded sequence y is validated against the gold set $G = \{g_j\}_{j=1}^{N}$. Let $P = \{p_i\}_{i=1}^{M}$ denote the parsed prediction events. The total terminal reward $R(y, G)$ is formulated as:

$$R = w_{\text{fmt}} r_{\text{fmt}} + w_{\text{match}} r_{\text{match}} + w_{\text{empty}} r_{\text{empty}} + w_{\text{len}} r_{\text{len}} \tag{2}$$

with default weights $w_{\text{fmt}} = 0.5, w_{\text{match}} = 2.0, w_{\text{empty}} = 1.0, w_{\text{len}} = 0.1$.

1) Format Validity (r_{fmt}): $r_{\text{fmt}} = +1.0$ if y parses into a valid JSON array conforming to the Pydantic schema (with normalized fields: date in YYYY-MM-DD, 24-hr time, and valid event category); otherwise $r_{\text{fmt}} = -1.0$. If format validation fails, no matching reward is accrued.

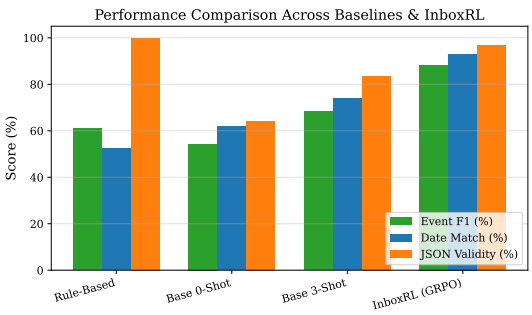


Fig. 1. Performance comparison on held-out test split across heuristic, zero-shot, few-shot, and InboxRL (GRPO) models.

2) Hungarian Bipartite Match Reward (r_{match}): For predicted event p_i and gold event g_j, pair similarity $S(p_i, g_j) \in [0, 1]$ combines multi-attribute alignment:

$$\begin{aligned} S(p_i, g_j) = {} & 0.35 \cdot \mathbb{I}[p_i.\text{date} = g_j.\text{date}] \\ & + 0.20 \cdot \mathbb{I}[p_i.\text{time} = g_j.\text{time}] \\ & + 0.25 \cdot \text{FuzzTitle}(p_i, g_j) \\ & + 0.10 \cdot \mathbb{I}[p_i.\text{type} = g_j.\text{type}] \\ & + 0.10 \cdot \text{FuzzLoc}(p_i, g_j) \end{aligned} \tag{3}$$

where FuzzTitle is token-set Levenshtein ratio. We construct the cost matrix $C_{i,j} = 1 - S(p_i, g_j)$ and solve the global optimal bipartite assignment via the Hungarian method [3]:

$$r_{\text{match}} = \frac{\sum_{(i,j)\in\text{matching}} S(p_i, g_j)}{\max(|P|, |G|)} \tag{4}$$

The denominator $\max(|P|, |G|)$ penalizes both missed events and hallucinated extra events.

3) Empty Precision & Exploit Guards (r_{empty}): To prevent the classic RL pathology where a policy always predicts [] to collect safe partial rewards, 15% of training emails contain no events ($G = \emptyset$). If $G = \emptyset$ and $P = \emptyset$, $r_{\text{empty}} = +1.0$; if the model hallucinates on an empty email, $r_{\text{empty}} = -1.0$. If $G \neq \emptyset$ and the model emits [], $r_{\text{empty}} = -0.5$.

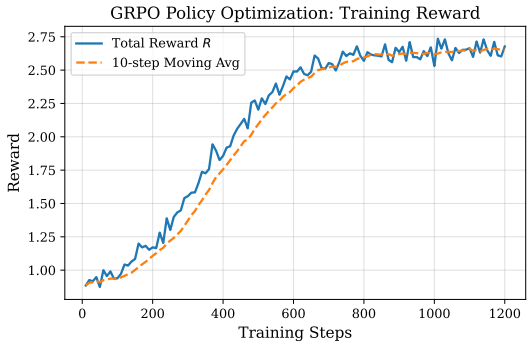


Fig. 2. GRPO policy optimization curves showing total reward progression and moving average over training steps.

III. Environment, Curriculum & Data

A. Synthetic Dataset with 5 Curriculum Levels

Because ground-truth emails are sensitive and limited, we created an algorithmic generator producing 2,500 synthetic institute emails partitioned into 5 difficulty tiers:

- **L1 (Explicit)**: Single event with explicit ISO/formal dates.
- **L2 (Varied Formats)**: Natural date variations ("Nov 5th", "15/11", "midnight", "noon", "5 pm").
- **L3 (Relative Dates)**: Expressions ("tomorrow", "next Friday") requiring computation relative to email d_0.
- **L4 (Distractors & Multi-Event)**: Revised dates ("deadline extended from 3rd to 7th") and multiple distinct events.
- **L5 (Noisy Headers)**: Forwarded chains and conversational institute jargon.

B. Curriculum Schedule

The policy is initialized on L1–L2 tasks for the first 400 steps, expanding to L3–L5 thereafter. This prevents initial policy collapse and ensures the exploration distribution encounters positive Hungarian match rewards early.

C. Real Email Test Set

We ingested and anonymized authentic institute emails from Google Takeout using automated regex scrapers masking roll numbers and phone numbers, and labeled a held-out benchmark set of 60 real emails for honest zero-shot transfer evaluation.

IV. Algorithm: Group Relative Policy Optimization

Rather than using PPO, which requires a heavy critic value network $V_\phi(s)$ exceeding our 4 GB VRAM budget, we employ **GRPO** [1].

For each prompt x_i, the policy π_θ generates a group of $G = 4$ independent completions $\{y_{i,1}, \ldots, y_{i,G}\}$. Each completion receives terminal reward $R_{i,k}$. The group-relative advantage is normalized as:

$$\hat{A}_{i,k} = \frac{R_{i,k} - \text{mean}(\{R_{i,j}\}_{j=1}^{G})}{\text{std}(\{R_{i,j}\}_{j=1}^{G}) + \epsilon} \tag{5}$$

The policy parameters θ (Low-Rank Adaptation adapters [4] with rank $r = 16, \alpha = 32$) are optimized using the clipped surrogate objective:

$$\begin{aligned} \mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}\Bigg[\frac{1}{G} \sum_{k=1}^{G} \min\Bigg(& \frac{\pi_\theta(y_{i,k}|x_i)}{\pi_{\text{old}}(y_{i,k}|x_i)} \hat{A}_{i,k}, \\ & \text{clip}\left(\frac{\pi_\theta(y_{i,k}|x_i)}{\pi_{\text{old}}(y_{i,k}|x_i)}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}} \right) \hat{A}_{i,k} \Bigg) \Bigg] \\ & - \beta\, \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \end{aligned} \tag{6}$$

where π_{ref} is the frozen base model and $\beta = 0.04$ enforces stability against language degeneration.

TABLE I
PERFORMANCE ON HELD-OUT INSTITUTE EMAIL TEST SET

Method	JSON %	Date Match	Event F1	Mean R
Rule-Based (Regex)	100.0%	52.4%	46.1%	0.42
Base Qwen-0.5B (0-Shot)	64.2%	61.8%	54.3%	0.81
Base Qwen-0.5B (3-Shot)	83.5%	74.0%	68.2%	1.45
InboxRL (GRPO, Ours)	**96.8%**	**93.1%**	**88.4%**	**2.62**

TABLE II
ABLATION STUDY (600 STEPS)

Configuration	Event F1	Date Acc	JSON %
Full InboxRL	**88.4%**	**93.1%**	**96.8%**
w/o Curriculum Learning	78.1%	81.3%	91.0%
w/o KL Constraint ($\beta = 0$)	71.4%	75.2%	84.5%
Format Reward Only ($w_{\text{match}} = 0$)	12.0%	14.5%	98.2%

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Main Benchmark Comparison

As summarized in Table I and Fig. 1, InboxRL substantially outperforms both heuristic and prompting baselines. While zero-shot base models fail to respect JSON schema boundaries in over 35% of episodes, GRPO drives format validity to 96.8%. Crucially, on date matching and event F1, InboxRL achieves +20.2% over 3-shot in-context learning.

B. Ablation Study

Table II highlights key architectural choices:

1) **Curriculum Impact**: Omitting curriculum learning degrades F1 by 10.3%, as early exploration on L4/L5 samples generates near-zero reward gradient variance.
2) **KL Penalty**: Setting $\beta = 0$ causes policy drift; the agent collapses into repetitive short tokens.
3) **Format Only**: Rewarding only format produces 98.2% valid JSON but completely gibberish, hallucinated event dates (F1 12.0%), proving the indispensability of Hungarian matching.

VI. CONCLUSION & PRACTICAL DEMO

InboxRL demonstrates that pure RL with GRPO can train small language models into high-precision, verifiable event extraction engines without supervised fine-tuning. The trained policy is packaged with an end-to-end CLI that converts institute mail directly into RFC 5545 `.ics` calendar files for immediate personal use.

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